Partial Derivatives and the Gradient

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: First and second partial derivatives**

For $f(x, y) = x^3y^2 - 4xy + y$, find:

a $f_x = 3x^2y^2 - 4y$

b $f_y = 2x^3y - 4x + 1$

c $f_{xx} = 6xy^2$

d $f_{xy} = 6x^2y - 4$

(a) $= 3x^2y^2 - 4y$; (b) $= 2x^3y - 4x + 1$; (c) $= 6xy^2$; (d) $= 6x^2y - 4$

2 **[Foundational]** • **Target: First and second partial derivatives**

For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.

$f_x = ye^{xy}\sin(x + y) + e^{xy}\cos(x + y)$.

3 **[Foundational]** • **Target: The gradient and directional derivatives**

Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.

$\nabla f = \langle 2xy, x^2 + 9y^2 \rangle$. At $(1, 2)$: $\nabla f(1, 2) = \langle 4, 37 \rangle$.

4 **[Foundational]** • **Target: The gradient and directional derivatives**

Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

$\nabla f = \langle 2x, 2y \rangle$, so $\nabla f(1, 1) = \langle 2, 2 \rangle$. Unit vector: $\hat{\mathbf{v}} = \langle \frac{3}{5}, \frac{4}{5} \rangle$. $D_{\mathbf{v}}f = \langle 2, 2 \rangle \cdot \langle \frac{3}{5}, \frac{4}{5} \rangle = \frac{6}{5} + \frac{8}{5} = \frac{14}{5}$.

5 **[Foundational]** • **Target: Tangent planes**

Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

$f_x = 2x$, $f_y = 2y$: at $(1, 2)$, $f_x = 2$, $f_y = 4$. Tangent plane: $z - 5 = 2(x - 1) + 4(y - 2)$, i.e. $z = 2x + 4y - 5$.

6 **[Foundational]** • **Target: Tangent planes**

Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.

$f_x = y$, $f_y = x$: at $(2, 3)$, $f_x = 3$, $f_y = 2$. Tangent plane: $z - 6 = 3(x - 2) + 2(y - 3)$, i.e. $z = 3x + 2y - 6$.

7 **[Foundational]** • **Target: Critical points of multivariable functions**

Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.

$f_x = 2x - 4 = 0$ gives $x = 2$; $f_y = 2y + 2 = 0$ gives $y = -1$. Critical point: $(2, -1)$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Critical points of multivariable functions**

Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

$f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$: $D = 2(2) - 0 = 4 > 0$ and $f_{xx} = 2 > 0$, so $(2, -1)$ is a local minimum.

9 **[Intermediate]** • **Target: The chain rule for multivariable functions**

Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.

$\frac{\partial z}{\partial x} = 2xy$, $\frac{\partial z}{\partial y} = x^2$; $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$. At $t = 1$: $x = 1, y = 1$. $\frac{dz}{dt} = 2(1)(1)(2) + 1(3) = 4 + 3 = 7$.

10 **[Intermediate]** • **Target: The chain rule for multivariable functions**

Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.

$\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = 2z$; $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$.
$\frac{dw}{dt} = 2x(-\sin t) + 2y\cos t + 2z = -2\cos t\sin t + 2\sin t\cos t + 2t = 2t$ (the first two terms cancel).

11 **[Intermediate]** • **Target: Maximum rate of increase**

Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.

$\nabla f = \langle 2x + y, x \rangle$; at $(1, 2)$: $\nabla f = \langle 4, 1 \rangle$. Direction of steepest increase: $\langle 4, 1 \rangle$ (or its unit vector). Maximum rate of increase: $|\nabla f| = \sqrt{16 + 1} = \sqrt{17}$.

12 **[Intermediate]** • **Target: Maximum rate of increase**

Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.

$\nabla f = \langle 2x, 2y \rangle$; at $(3, 4)$: $\langle 6, 8 \rangle$. Steepest descent is the opposite direction: $\langle -6, -8 \rangle$ (pointing toward the origin, the minimum).

13 **[Intermediate]** • **Target: Implicit partial differentiation**

For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).

$2x + 2z\frac{\partial z}{\partial x} = 0$, so $\frac{\partial z}{\partial x} = -\frac{x}{z}$.

14 **[Intermediate]** • **Target: Classifying critical points with saddle points**

Find and classify the critical point(s) of $f(x, y) = x^2 - y^2$, using the second derivative test.

$f_x = 2x = 0$, $f_y = -2y = 0$ gives $(0, 0)$. $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$: $D = (2)(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point (neither a local max nor min).

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Classifying critical points with saddle points** • **Cross-Lesson: 8.02 Volumes of Revolution**

Find and classify the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

$f_x = 3x^2 - 3y = 0$ gives $y = x^2$. $f_y = -3x + 3y^2 = 0$ gives $x = y^2 = x^4$, so $x^4 - x = 0$, $x(x^3 - 1) = 0$: $x = 0$ or $x = 1$. Critical points $(0, 0)$ and $(1, 1)$. $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$. At $(0, 0)$: $D = 0 - 9 = -9 < 0$: saddle point. At $(1, 1)$: $D = (6)(6) - 9 = 27 > 0$ and $f_{xx} = 6 > 0$: local minimum.

16 **[Challenging]** • **Target: An applied optimization problem** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

A company's profit is modeled by $P(x, y) = 100x + 120y - x^2 - 2y^2 - xy$, where x and y are units of two products. Find the production levels that maximize profit by setting $P_x = 0$ and $P_y = 0$.

$P_x = 100 - 2x - y = 0$; $P_y = 120 - 4y - x = 0$. From the first: $y = 100 - 2x$. Substituting: $120 - 4(100 - 2x) - x = 0$: $120 - 400 + 8x - x = 0$: $7x = 280$, $x = 40$. Then $y = 100 - 80 = 20$. Maximum profit at $(x, y) = (40, 20)$.

17 **[Challenging]** • **Target: Higher-order mixed partial derivatives** • **Cross-Lesson: 9.01 Separable Differential Equations**

For $f(x, y) = x^3y^2 + \sin(xy)$, find f_{yx} and f_{xy}, and verify they are equal (as guaranteed by Clairaut's theorem for continuous mixed partials).

$f_x = 3x^2y^2 + y\cos(xy)$: $f_{xy} = 6x^2y + \cos(xy) - xy\sin(xy)$. $f_y = 2x^3y + x\cos(xy)$: $f_{yx} = 6x^2y + \cos(xy) - xy\sin(xy)$. Both equal, confirming Clairaut's theorem.

18 **[Challenging]** • **Target: Linear approximation in two variables** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

Use the linear approximation $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ to estimate $f(2.1, 2.9)$ for $f(x, y) = x^2y$, using the point $(2, 3)$.

$f(2, 3) = 12$. $f_x = 2xy$: $f_x(2, 3) = 12$. $f_y = x^2$: $f_y(2, 3) = 4$. Estimate: $12 + 12(0.1) + 4(-0.1) = 12 + 1.2 - 0.4 = 12.8$.

19 **[Challenging]** • **Target: Linear approximation in two variables** • **Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

The volume of a cylinder is $V(r, h) = \pi r^2 h$. A cylindrical can is manufactured with radius $r = 3$ cm and height $h = 10$ cm, but due to manufacturing tolerances, the radius might be off by as much as 0.05 cm and the height by as much as 0.1 cm. Use the total differential $dV = V_r\,dr + V_h\,dh$ to estimate the maximum possible error in the computed volume.

$V_r = 2\pi rh$: at $(3, 10)$, $V_r = 60\pi$. $V_h = \pi r^2$: at $(3, 10)$, $V_h = 9\pi$. Maximum error estimate: $|dV| \le |V_r||dr| + |V_h||dh| = 60\pi(0.05) + 9\pi(0.1) = 3\pi + 0.9\pi = 3.9\pi \approx 12.25$ cm^3.

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20 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 11.01 Sequences and Series

For $f(x, y) = x^2y - 3xy^2 + y^3$, find each of the following at the point $(1, -1)$:

a $f_x(1, -1)$ $f_x = 2xy - 3y^2$: $f_x(1, -1) = 2(1)(-1) - 3(1) = -2 - 3 = -5$.

b $f_y(1, -1)$ $f_y = x^2 - 6xy + 3y^2$: $f_y(1, -1) = 1 - 6(1)(-1) + 3(1) = 1 + 6 + 3 = 10$.

c The gradient vector $\nabla f(1, -1)$ $\nabla f(1, -1) = \langle -5, 10 \rangle$.

d The directional derivative in the direction of $\langle 1, 0 \rangle$ $D_{\langle 1, 0 \rangle} f = \nabla f \cdot \langle 1, 0 \rangle = -5$ (since $\langle 1, 0 \rangle$ is already a unit vector).

(a) $f_x = 2xy - 3y^2$: $f_x(1, -1) = 2(1)(-1) - 3(1) = -2 - 3 = -5$.; (b) $f_y = x^2 - 6xy + 3y^2$: $f_y(1, -1) = 1 - 6(1)(-1) + 3(1) = 1 + 6 + 3 = 10$.; (c) $\nabla f(1, -1) = \langle -5, 10 \rangle$.; (d) $D_{\langle 1, 0 \rangle} f = \nabla f \cdot \langle 1, 0 \rangle = -5$ (since $\langle 1, 0 \rangle$ is already a unit vector).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: First-Order Partial Derivatives

Find f_x and f_y for $f(x, y) = 3x^2y - 5xy^3 + 4x - 2y$.

a. $f_x = 6xy - 5y^3 + 4$, $f_y = 3x^2 - 15xy^2 - 2$ ✓

b. $f_x = 6xy - 5y^3$, $f_y = 3x^2 - 15xy^2$

c. $f_x = 6x - 5y^3 + 4$, $f_y = 3x^2 - 15y^2 - 2$

d. $f_x = 3x^2 - 15xy^2$, $f_y = 6xy - 5y^3$

Treat y as constant: $f_x = 6xy - 5y^3 + 4$. Treat x as constant: $f_y = 3x^2 - 15xy^2 - 2$.

22 [Foundational] • Target: First-Order Partial Derivatives

Find $\frac{\partial z}{\partial x}$ for $z = e^{x^2y}$.

a. $2xye^{x^2y}$ ✓

b. $x^2e^{x^2y}$

c. $2xe^{x^2y}$

d. $2xye^{2xy}$

By the chain rule: $\frac{\partial z}{\partial x} = e^{x^2y} \frac{\partial}{\partial x}(x^2y) = 2xye^{x^2y}$.

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23 [Foundational] • Target: First-Order Partial Derivatives

Find $f_y(1, 0)$ for $f(x, y) = \ln(x^2 + y^2 + 1)$.

a. 0 ✓

b. 1

c. 2

d. $1/2$

$f_y = \frac{2y}{x^2+y^2+1}$. Evaluating at $(1, 0)$: $f_y(1, 0) = \frac{0}{1+0+1} = 0$.

24 [Foundational] • Target: Second-Order Partials and Clairaut's Theorem

What does Clairaut's Theorem (equality of mixed partials) state for continuous second partial derivatives?

a. $f_{xy} = f_{yx}$ ✓

b. $f_{xx} = f_{yy}$

c. $f_{xy} = -f_{yx}$

d. $f_{xx} + f_{yy} = 0$

If the mixed partial derivatives f_{xy} and f_{yx} are both continuous on an open set, they are equal: $f_{xy} = f_{yx}$.

25 [Foundational] • Target: Second-Order Partials and Clairaut's Theorem

Find f_{xx} for $f(x, y) = x^4y^2 - 2x^2y$.

a. $12x^2y^2 - 4y$ ✓

b. $4x^3y^2 - 4xy$

c. $2x^4 - 0$

d. $12x^2 - 4$

$f_x = 4x^3y^2 - 4xy$. Differentiating with respect to x again: $f_{xx} = 12x^2y^2 - 4y$.

26 [Foundational] • Target: Second-Order Partials and Clairaut's Theorem

Verify whether $u(x, y) = e^x \sin y$ satisfies Laplace's equation $u_{xx} + u_{yy} = 0$.

a. **Yes,** $u_{xx} + u_{yy} = e^x \sin y - e^x \sin y = 0$ ✓

b. No, $u_{xx} + u_{yy} = 2e^x \sin y$

c. No, it equals $e^x \cos y$

d. Only when $x = 0$

$u_x = e^x \sin y \Rightarrow u_{xx} = e^x \sin y$. $u_y = e^x \cos y \Rightarrow u_{yy} = -e^x \sin y$. $u_{xx} + u_{yy} = 0$.

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27 [Foundational] • Target: The Gradient Vector and Directional Derivatives

Find the gradient $\nabla f(x, y)$ for $f(x, y) = x^2y + 3y^2$ at the point $(2, 1)$.

a. $\langle 4, 10 \rangle$ ✓

b. $\langle 4, 7 \rangle$

c. $\langle 2, 6 \rangle$

d. $\langle 10, 4 \rangle$

$\nabla f = \langle 2xy, x^2 + 6y \rangle$. At $(2, 1)$: $\nabla f(2, 1) = \langle 2(2)(1), 4 + 6(1) \rangle = \langle 4, 10 \rangle$.

28 [Foundational] • Target: The Gradient Vector and Directional Derivatives

In which direction does a differentiable function f increase most rapidly at a point P?

a. **In the direction of the gradient vector** $\nabla f(P)$ ✓

b. Opposite to $\nabla f(P)$

c. Orthogonal to $\nabla f(P)$

d. Along the level curve

The directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = ||\nabla f|| \cos\theta$ is maximized when $\cos\theta = 1$, which is the direction of ∇f.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Gradient Vector and Directional Derivatives

Find the directional derivative of $f(x, y) = 2x^2 - 3y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

a. $-12/5$ ✓

b. $12/5$

c. -24

d. 0

$\nabla f = \langle 4x, -6y \rangle \Rightarrow \nabla f(1, 1) = \langle 4, -6 \rangle$. Unit vector $\mathbf{u} = \langle 3/5, 4/5 \rangle$.
$D_{\mathbf{u}} f = \langle 4, -6 \rangle \cdot \langle 3/5, 4/5 \rangle = 12/5 - 24/5 = -12/5$.

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30 [Intermediate] • Target: Tangent Planes and Linear Approximations

Find the equation of the tangent plane to $z = x^2 + y^2$ at the point $(1, 2, 5)$.

a. $z - 5 = 2(x-1) + 4(y-2)$ ✓

b. $z - 5 = (x-1) + 2(y-2)$

c. $2x + 4y - z = 0$

d. $z = 2x + 4y + 5$

$z_x = 2x \Rightarrow z_x(1,2) = 2$. $z_y = 2y \Rightarrow z_y(1,2) = 4$. Plane: $z - 5 = 2(x-1) + 4(y-2) \Rightarrow 2x + 4y - z = 5$.

31 [Intermediate] • Target: Tangent Planes and Linear Approximations

Use the linear approximation $L(x, y)$ of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$ to estimate $f(3.02, 3.97)$.

a. 4.988 ✓

b. 5.012

c. 5.000

d. 4.950

$f(3,4) = 5$. $f_x = x/\sqrt{x^2+y^2} = 3/5 = 0.6$; $f_y = 4/5 = 0.8$.
$L = 5 + 0.6(0.02) + 0.8(-0.03) = 5 + 0.012 - 0.024 = 4.988$.

32 [Intermediate] • Target: Tangent Planes and Linear Approximations

Find the normal vector to the level surface $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.

a. $\langle 2, 4, 4 \rangle$

b. $\langle 1, 2, 2 \rangle$

c. **Both $\langle 1, 2, 2 \rangle$ and $\langle 2, 4, 4 \rangle$ are valid normal vectors** ✓

d. $\langle 2, 2, 1 \rangle$

The gradient of $F(x, y, z) = x^2 + y^2 + z^2$ is $\nabla F = \langle 2x, 2y, 2z \rangle = \langle 2, 4, 4 \rangle = 2\langle 1, 2, 2 \rangle$, which is normal to the tangent plane.

33 [Intermediate] • Target: Second-Order Partials and Clairaut's Theorem

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. **Lagrange multiplier** ✓

b. eigenvalue

c. gradient

d. curvature

Applying Application of Derivatives principles, λ = Lagrange multiplier. Therefore, the verified correct option is Lagrange multiplier.

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34 [Intermediate] • Target: The Gradient Vector and Directional Derivatives

Second-order Taylor approximation:

$f(x,y) \approx f(a,b) + f_x(x-a) + f_y(y-b) + \frac{1}{2}[f_{xx}(x-a)^2 + 2f_{xy}(x-a)(y-b) + f_{yy}(y-b)^2]$. This involves:

a. only first partials

b. only second partials

c. third partials

d. **both first and second partials ✓**

Second-order Taylor uses both first and second partial derivatives.

35 [Intermediate] • Target: Tangent Planes and Linear Approximations

Chain rule with two paths: $z = f(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x\cos\theta + f_y\sin\theta$ ✓

b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

Chain rule: $z_r = f_x x_r + f_y y_r = f_x\cos\theta + f_y\sin\theta$.

36 [Intermediate] • Target: First and second partial derivatives

Classify the critical points of $f(x,y) = x^2 - 4x + y^3 - 12y$:

a. $(2,2)$: saddle

b. $(2,2)$ min; $(2,-2)$ saddle

c. $(2,2)$**: local min;** $(2,-2)$**: local max** ✓

d. $(2,2)$ saddle; $(2,-2)$ saddle

$f_x = 2x - 4 = 0 \Rightarrow x = 2$; $f_y = 3y^2 - 12 = 0 \Rightarrow y = \pm 2$. $H = (2, 6y, 0)$: at $(2,2)$: $D = 2 \cdot 12 > 0$, $f_{xx} = 2 > 0 \Rightarrow$ min; at $(2,-2)$: $D = 2 \cdot (-12) < 0 \Rightarrow$ saddle.

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37 [Intermediate] • Target: The gradient and directional derivatives

Chain rule: $z = f(x, y)$, $x = x(t)$, $y = y(t)$. Then $dz/dt =$

a. $f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$ ✓

b. $f_y dy/dt$

c. $f_x dx/dt$

d. $f_{xx} + f_{yy}$

Applying Partial Derivatives and Gradient principles, Multivariable chain rule. Therefore, the verified correct option is $f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Tangent planes • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find all critical points of $f(x, y) = x^3 + y^3 - 3x - 3y$.

a. $(\pm 1, \pm 1)$ ✓

b. $(0, 0)$ only

c. $(1, 1)$ only

d. $(1, -1)$ and $(-1, 1)$ only

Setting $f_x = 3x^2 - 3 = 0$ and $f_y = 3y^2 - 3 = 0$ yields $x = \pm 1$ and $y = \pm 1$, giving the four critical points $(\pm 1, \pm 1)$.

39 [Challenging] • Target: Critical points of multivariable functions • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Implicit function theorem: at (a, b), $\partial y/\partial x =$

a. $D_x F$

b. $-(D_y F)^{-1} D_x F$ ✓

c. 0

d. I

Applying Partial Derivatives and Gradient principles, IFT derivative formula. Therefore, the verified correct option is $-(D_y F)^{-1} D_x F$.

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40 [Challenging] • Target: The chain rule for multivariable functions • Cross-Lesson: 11.01 Sequences and Series

Second mixed partial: for smooth f, $f_{xy} =$

a. f_{yx} ✓

b. f_{yy}

c. f_{xx}

d. 0

Applying Partial Derivatives and Gradient principles, Clairaut: $f_{xy} = f_{yx}$. Therefore, the verified correct option is f_{yx}.

41 [Challenging] • Target: Maximum rate of increase • Cross-Lesson: 12.01 Convergence Tests

Euler's theorem for homogeneous functions: if $f(tx, ty) = t^n f(x, y)$, then $xf_x + yf_y =$

a. f

b. nf ✓

c. $n(n-1)f$

d. 0

Applying Partial Derivatives and Gradient principles, Euler: $xf_x + yf_y = nf$. Therefore, the verified correct option is nf.

42 [Challenging] • Target: Implicit partial differentiation • Cross-Lesson: 12.02 Power Series and Taylor Series

What is the maximum rate of change of $f(x, y) = x^3 - 3xy^2 - y$ at $(2,\ 1)$?

a. $5\sqrt{10}$ ✓

b. 250

c. 22

d. 13

The maximum rate is $||ablaf||$. Here $ablaf(2,\ 1) = \langle 9,\ -13\rangle$, so it is $5\sqrt{10}$.

43 [Challenging] • Target: Classifying critical points with saddle points • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

What is the maximum rate of change of $f(x, y) = x^2 - 2xy - 3y^2$ at $(3,\ -2)$?

a. 136

b. $2\sqrt{34}$ ✓

c. 16

d. 10

The maximum rate is $||ablaf||$. Here $ablaf(3,\ -2) = \langle 10, 6\rangle$, so it is $2\sqrt{34}$.

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44 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 15.01 Double Integrals

What is the maximum rate of change of $f(x, y) = 3x^3 + xy^2$ at $(2,\ 1)$?

a. 1385

b. $\sqrt{1385}$ ✓

c. 41

d. 37

The maximum rate is $||ablaf||$. Here $ablaf(2,\ 1) = \langle 37, 4\rangle$, so it is $\sqrt{1385}$.

45 [Challenging] • Target: Higher-order mixed partial derivatives • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Local extrema of f can occur only at critical points (interior) or on the:

a. axes

b. level curves

c. **boundary** ✓

d. Hessian eigenvalues

Global extrema on closed bounded region: check interior critical points and boundary.

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